

Md. Rubaet Kabir Zishan

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EDUCATION

B.Sc. in Computer Science & Engineering

May 2022 – May 2026 (*Expected*)

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

- **Cumulative GPA:** 3.24 / 4.00 (Final semester in progress)
- **Thesis:** *Bangla ↔ Chakma Neural Machine Translation via Fine-Tuned T5* — bidirectional seq2seq fine-tuning for two linguistically distinct, extremely low-resource languages with minimal parallel corpora.
- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Algorithms & Complexity, Software Engineering, Computer Networks, Database Systems, Probability & Statistics.

RESEARCH INTERESTS

Applied Machine Learning	Large Language Models	Agentic & Multi-Agent AI
Explainable AI (XAI)	Reinforcement Learning	AI for Environment & Health
Natural Language Processing	Probabilistic Modeling	Low-Resource Language Tech
AI Security & Cybersecurity		

PUBLICATIONS

- **“AI Integrated Automated Door Lock System Through Face Recognition.”** *IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, 2025. [Published — IEEE Xplore]
- **“Chashi-IoT: Developing a Sustainable Integrated Agricultural IoT System Using Cisco Packet Tracer.”** *IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE)*, 2025. [Published — IEEE Xplore]
- **“Bangla ↔ Chakma Neural Machine Translation via Fine-Tuned T5.”** *Undergraduate Thesis — Target: ACL / EMNLP Workshop or NLP Journal*, 2026. [In Preparation]

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher

Jan 2025 – Present

Dept. of Computer Science & Engineering, MIST, Dhaka, Bangladesh

Supervisor: Lt Col Muhammad Nazrul Islam, PhD • Associate Professor, CSE — MIST • nazrul@cse.mist.ac.bd
Top 2% Scientist globally (Stanford-Elsevier 2022–2025) • 69+ Q1/Q2 journal articles • USD 250,000+ in research grants

- Co-authored two IEEE conference papers accepted prior to graduation: *Automated Door Lock System Using Face Recognition* (IEEE QPAIN 2025, **Published**) and *Development of a Sustainable Integrated Agricultural IoT System* (IEEE WIECON-ECE 2025, **Published**).
- Designing task-adaptive fine-tuning protocols for Bangla↔Chakma translation using the T5 framework; building BLEU/chrF++ evaluation pipelines and a web deployment interface for Chakma-speaking communities.
- Contributing to experimental design, methodology discussions, and manuscript preparation within an active ML/NLP research group.

RESEARCH & TECHNICAL PROJECTS

Bangla ↔ Chakma Neural Machine Translation [Undergraduate Thesis]

2025 – Present

Python • PyTorch • Hugging Face • T5/mT5 • Seq2Seq • BLEU/chrF++ Evaluation

- Fine-tuning a Bengali T5 model for bidirectional Chakma↔Bengali translation, targeting two linguistically distinct low-resource languages with very limited parallel corpora — Chakma is an indigenous language spoken by roughly 400,000 people with almost no digital NLP resources.
- Benchmarking against mBART and mT5 baselines using BLEU, chrF++, and human preference evaluation to measure the transfer-learning gap in this language pair.
- Deploying as a web application to validate end-to-end system performance and give Chakma speakers direct access to the model.

Churn Sentinel — Explainable AI Customer Retention System [GitHub]

2026

Python • XGBoost • SHAP • LangChain • Multi-Agent AI

- Trained an XGBoost classifier with stratified cross-validation and Optuna hyperparameter search on class-imbalanced telecom churn data.
- Integrated SHAP for global (beeswarm, summary bar) and local (force plot, waterfall) explanations to make model predictions auditable at the individual decision level.
- Built a LangChain multi-agent pipeline (Behavior Detector → Discount Evaluator → Personalized Outreach Generator) that converts SHAP risk scores into autonomous, context-aware retention actions.

Adaptive Traffic Signal Control via Reinforcement Learning [GitHub]

2025

Python • PPO • Stable-Baselines3 • SUMO Simulator • OpenAI Gym

- Built a custom Gym-compatible SUMO environment with normalized per-lane state features and a composite reward combining queue-length penalty and vehicle-throughput bonus.
- Enforced safe exploration through yellow-phase protection constraints, preventing conflicting signal transitions during agent training.
- Trained a PPO agent for 200,000 steps with EvalCallback checkpointing and TensorBoard logging; the learned policy reduced average queue length relative to a fixed-cycle baseline across all evaluation episodes.

Bayesian Water Quality Analysis — Bangladesh [GitHub]

2025

Python • PyMC • Bayesian Hierarchical Modeling • Environmental Data Science

- Applied Bayesian hierarchical models to quantify uncertainty in water quality measurements across geographically distributed sampling sites in Bangladesh, where chronic arsenic exposure affects an estimated 20 million people.
- Derived posterior predictive distributions for arsenic, dissolved solids, and microbial thresholds, producing probabilistic risk estimates rather than binary pass/fail classifications.
- Dataset and methodology contribute to the evidence base for policy decisions on rural water safety in South Asia.

Genomic Variant Classifier — Pathogenicity Prediction [GitHub]

2025

Python • Scikit-Learn • Random Forest • Bioinformatics

- Built a multi-class Random Forest classifier to predict SNV pathogenicity (benign / pathogenic / variant of uncertain significance) using ClinVar-annotated variants with in-silico features including SIFT, PolyPhen-2, and allele frequency scores.
- Addressed severe class imbalance between pathogenic and benign variants through stratified sampling and threshold tuning; evaluated using precision, recall, F1, and ROC-AUC per class.
- Motivated by the clinical cost of misclassification: a variant incorrectly labeled benign can delay diagnosis in hereditary cancer and cardiac conditions where early intervention is decisive.

PROFESSIONAL EXPERIENCE

Software Engineering Intern — Aurify (Part-time, Remote)

Aug 2025 – Apr 2026

Chicago, IL, USA • Node.js / React SaaS Platform

- Designed Playwright / JavaScript E2E test suites (Page Object Model) across critical user flows and developed a structured defect triage process that systematically reduced regression escapes over three consecutive production release cycles — applying the same hypothesis-driven thinking used in experimental research to software quality problems.
- Diagnosed full-stack bugs in a Node.js / React codebase, validated REST API contracts via Postman, and produced 30+ reproducible defect reports with environment-specific reproduction steps — work that required the same precision and documentation discipline expected in research settings.

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, SQL, LaTeX
ML & Deep Learning	Scikit-Learn, XGBoost, PyTorch, Hugging Face Transformers, T5 / BERT / mT5
Explainability	Fine-tuning, Stable-Baselines3, LangChain
Probabilistic Methods	SHAP, LIME, feature attribution, model cards
Reinforcement Learning	Bayesian inference, PyMC, hierarchical modeling, posterior predictive checks
Research Tools	PPO, DQN, OpenAI Gym, SUMO simulator
Software Engineering	Jupyter, MATLAB, Git / GitHub, LaTeX, Overleaf, Weights & Biases
	React, Node.js, Playwright, PostgreSQL, REST APIs

COMPETITIVE PROGRAMMING

300+ problems solved across LeetCode, Codeforces, and CodeChef, covering data structures, graph algorithms, dynamic programming, and combinatorics.

LeetCode: rubayetkabirzisan • Codeforces: mission27 • CodeChef: rubaet_kabir

STANDARDIZED TESTS

GRE General Test Preparation in progress; examination to be scheduled.

IELTS Academic Preparation in progress; examination to be scheduled.

REFERENCES

Lt Col Muhammad Nazrul Islam, PhD

Associate Professor, CSE
Military Institute of Science
& Technology (MIST)
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Thesis supervisor.

Top 2% Scientist

(Stanford-Elsevier 2022–25).

Dr. Md. Akhtaruzzaman

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Co-author, IEEE QPAIN 2025.

PhD, IIUM Malaysia.

Formerly, MIST CSE.

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Co-author, IEEE WIECON 2025.

PhD, Univ. of Dhaka.

Formerly, MIST CSE.